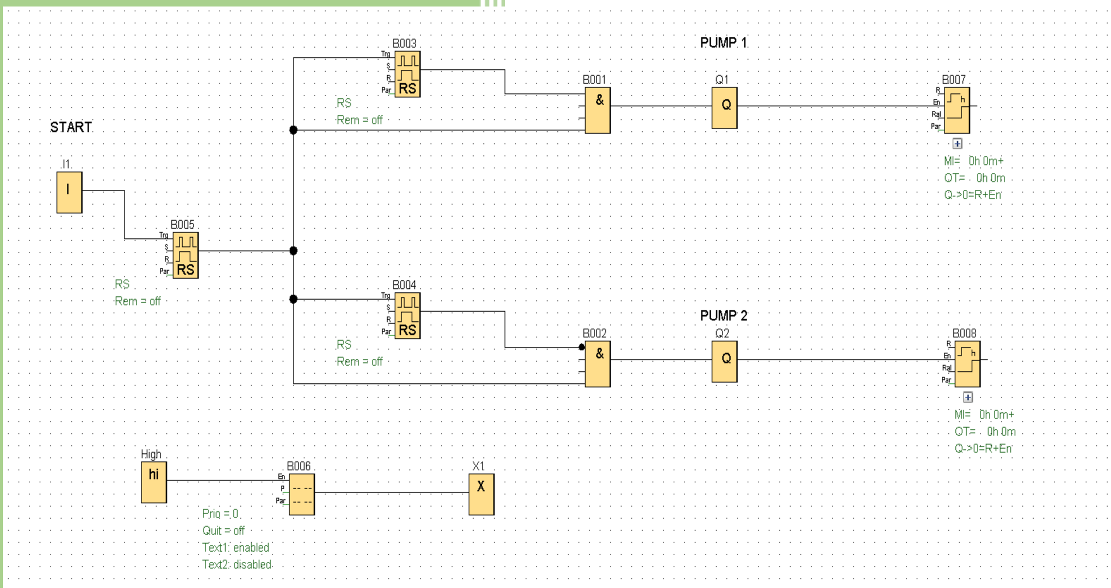
Pump Control System Using PLC



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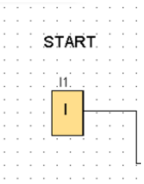
1. Introduction

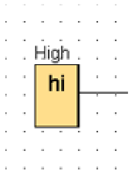
This project represents an automated control system for two water pumps using PLC. The system is designed to alternate operation between Pump 1 and Pump 2, monitor operating time, and display status information on the LOGO display.

The design ensures:

- Alternating pump operation
- Equal usage distribution
- Runtime monitoring
- Visual status indication

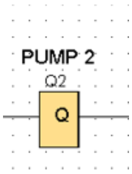
Inputs:

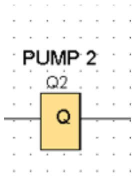




- START (I): System activation signal
- High Level Sensor (HI): Indicates tank full condition

Outputs:





- Q1: Pump 1 control
- Q2: Pump 2 control

Internal Blocks:

- RS Flip-Flops (B003, B004, B005)
- Message Display Block (B006)

3. Functional Description

When the system starts:

1. The START signal activates the control logic.
2. The system decides which pump should run using RS logic.
3. Only one pump operates at a time.
4. Each pump runtime is recorded.
5. The system displays pump status and total runtime.
6. If switching occurs, the other pump takes over.

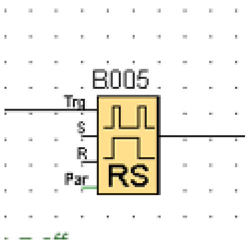
4. Block-by-Block Explanation

4.1 START Input (I)

- Function: Initiates system operation
- Type: Digital Input
- Behavior: When ON, enables system logic

4.2 RS Flip-Flop (B005)

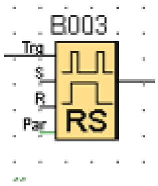
- Function: Main latch for system activation
- Inputs:
    - S (Set): From START input
    - R (Reset): Not used or external condition
- Output:
    - Enables both pump control branches



Working Principle:
Once START is pressed, the system remains active until reset.

4.3 RS Flip-Flops (B003 & B004)

- Function: Alternating control between Pump 1 and Pump 2



B003 (Pump 1 Control Memory)

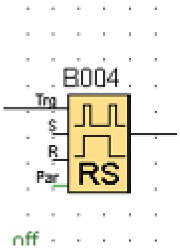
- Stores state for Pump 1

B004 (Pump 2 Control Memory)

- Stores state for Pump 2

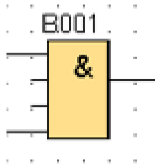
Working Principle:

- Only one flip-flop is active at a time
- They toggle between pumps
- Prevent simultaneous operation



4.4 AND Gate (B001) – Pump 1 Logic

- Inputs:
  - Output from B003 (Pump 1 enable)
  - System enable (from B005)
- Output:
  - Drives Q1 (Pump 1)

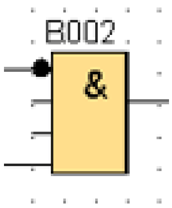


Logic:
Pump 1 operates only when:

- System is ON
- Pump 1 is selected

4.5 AND Gate (B002) – Pump 2 Logic

- Inputs:
  - Output from B004 (Pump 2 enable)
  - System enable
- Output:
  - Drives Q2 (Pump 2)



Logic:
Pump 2 operates only when:

- System is ON
- Pump 2 is selected

4.6 Output Blocks (Q1 & Q2)

- Q1: Controls Pump 1
- Q2: Controls Pump 2

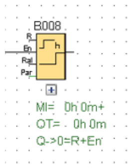
Behavior:

- Only one output is active at any time
- Directly connected to physical relays or contactors

4.7 Runtime Counters (B007 & B008)

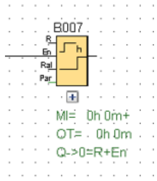
B007 – Pump 1 Timer

- Measures total runtime of Pump 1
- Starts counting when Q1 is ON



B008 – Pump 2 Timer

- Measures total runtime of Pump 2
- Starts counting when Q2 is ON



Displayed Parameters:

- OT (Operating Time)
- MI/MN (Minimum/Maximum values if configured)

4.8 Message Display Block (B006)

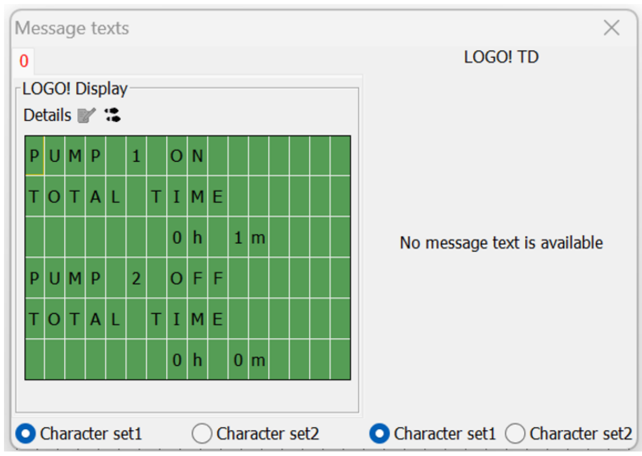
- Function: Displays system information on LOGO screen

Displayed Information:

- Pump 1 status (ON/OFF)
- Pump 1 total time
- Pump 2 status (ON/OFF)
- Pump 2 total time

Working Principle:

- Reads outputs (Q1, Q2)
- Reads timer values
- Updates display dynamically



- Function: Safety or control condition
- Can be used to:
    - Stop pumps when tank is full
    - Trigger alarms

5. System Operation Sequence

1. User presses START
2. System latch activates (B005)
3. First pump (Pump 1) starts
4. Runtime begins counting (B007)
5. After condition or switching logic:
    - Pump 1 stops
    - Pump 2 starts
6. Runtime for Pump 2 begins (B008)
7. Display updates continuously

6. Key Design Features

- Alternating pump logic (load balancing)
- Prevention of simultaneous pump operation
- Runtime monitoring for maintenance
- Visual feedback via display
- Expandable structure for additional sensors

7. Possible Improvements

- Add automatic switching based on time
- Add fault detection (overload, dry run)
- Integrate PID for level control
- Add SCADA or IoT monitoring

8. Conclusion

This project demonstrates a complete industrial control solution using PLC. It integrates logic control, timing, monitoring, and visualization in a structured and scalable design suitable for real-world water pumping systems.